

Review retrospective:
*the sharp covariance rate — parity vanishing verified, three
more “proof in progress” docstrings on completed proofs*

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Abstract

A soundness review of `Laplace/Multi/CovarianceSharp.lean` (4179 lines): the sharp-rate multivariate Laplace covariance, `gibbsCov_first_order_rate_sharp`, upgrading the weak $O(t^{-3/2})$ correction to the sharp $O(t^{-2})$ by odd-Gaussian parity vanishing. Result: a clean fidelity pass — the parity mechanism that drives the upgrade is correct — with three stale-status docstring fixes; the only simplification surface (a large batch of unused `set` equation-names) was deferred report-only.

1 Setting

The sharp half of the multivariate covariance result — the file the weak-track Covariance review cross-references. Where the weak track stops at $O(t^{-3/2})$ via triangle bounds, the sharp track subtracts the *vanishing* leading half-power correction (odd in u , killed against the even Gaussian weight) to reach $O(t^{-2})$.

2 What was reviewed

The headline `gibbsCov_first_order_rate_sharp`, the parity-vanishing lemmas (`integral_centered_bilinear_cubicJet_eq_zero`, `integral_dot_mul_quadJet_eq_zero`, the centering identity `integral_centered_bilinear_gaussianWeight_eq_zero`), the corrected-bracket rewrite `integral_centered_bilinear_eq_corrected_bracket` that is the actual weak→sharp upgrade step, and the sharp intermediate bounds.

3 Fidelity / soundness findings

Sound; the parity mechanism checks out. The reviewer confirmed the headline matches the primer’s sharp form — main term $\langle a, H^{-1}b \rangle$ (not H), positive sign, sharp K/t rate (covariance itself $O(t^{-2})$) — and, crucially, that *the upgrade from weak to sharp is genuinely driven by parity*: the two key integrands are odd $\times$ even and vanish, and the corrected-bracket identity is exactly the place where the weak triangle bound is replaced by subtracting the vanishing odd piece. “No wrong rate, no wrong contraction, no sign error.” (One benign naming quirk: a parity lemma’s jet parameter is named `q $\varphi$` where the role is `q $\psi$` ; the statement is correct.)

4 Simplifications and documentation fixes applied

Three stale-status docstrings. The sharp headline `gibbsCov_first_order_rate_sharp` still announced “(..., statement only)” and “Statement is locked in here; proof is in progress” — on the fully-proved headline of the file (the same false-status pattern the weak headline carried in the previous review). Two helper docstrings likewise said their work was “deferred” / that “sub-helpers will be added in subsequent stages,” though the proofs and helpers are present. All three corrected. Comment-only; build green, signatures byte-identical; the reviewer affirmed.

What was deferred. The file is already minimal on opens (one load-bearing `open MeasureTheory`, no dead `Module`, no redundant re-opens) and has no dead private lemmas. Its one real dead-code surface is a *large* batch — roughly thirty unused `set ...with hX` equation-names across eight lemmas. These are build-arbitrated-removable but low-value and high-count; consistent with the report-only call on the comparable batches in the `IntegralRemainder` and `Covariance` reviews, they were left to a dedicated hygiene tide rather than risk a thirty-name batch edit on a 4179-line file.

5 Disagreements and open questions

None. Two pre-existing unused-hypothesis-in-type warnings
(`abs_rescaledObservable_quadratic_error_le` here,
`abs_rescaledObservable_linear_error_le` in the `RescaledIntegrals` dependency) are
signature-level (removing the hypothesis is an API change), so report-only.

6 Lessons

- **“Proof in progress” on the headline is now a recurring pattern across the Multi/files.** Three consecutive large files (`RescaledIntegrals`, `Covariance`, `CovarianceSharp`) each carried a stale “statement only / in progress” or “deferred” note on work that has since completed — these were written headline-first during development and never refreshed. On a mature multi-stage file, the status lines are the highest-yield fidelity target.
- **Know when to defer a large dead-code batch.** A handful of dead locals is worth landing; thirty unused equation-names across eight lemmas on a 4k-line file is a hygiene-tide job, not a review-tide one — the value per edit is low and the batch-edit risk is real. Report it precisely (names + lemmas) so the sweep is mechanical when it happens.

7 Follow-ups

- Hygiene tide: sweep the 30 dead `set ...with hX` equation-names (listed in the review log) and the two unused-hypothesis-in-type parameters.
- The last never-reviewed laplace file is `Multi/CovarianceExplicit` (19611 lines) — genuinely sectioned-review territory, beyond a single-pass consult.